

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

DECEMBER/JANUARY EXAMINATION

AMIT3353 MOBILE APPLICATION DEVELOPMENT

WEDNESDAY, 7 JANUARY 2026

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMIT3353 MOBILE APPLICATION DEVELOPMENT

SME Bank has launched the ELSA (Enterprise Lifecycle Scoring Assessment) mobile app to support Malaysia's SME sector and strengthen the nation's digital transformation agenda as ASEAN Chair 2025. The AI-powered app aims to assist 10,000 SMEs by providing performance assessments, identifying business gaps, and connecting entrepreneurs to suitable advisory or financing programs. This mobile app builds on the earlier web-based ELSA framework introduced in 2021, which had supported more than 5,000 entrepreneurs through national collaborations.

ELSA offers a complete digital ecosystem by combining diagnostics, training, and financing in one platform. Through its integration with ModalNiaga and ScoreXcess, SMEs can submit financing applications more efficiently, benefit from faster scoring, and enjoy auto-filled forms based on assessment results. This reduces application barriers and speeds up decisions. Recognised with innovation awards, the ELSA app demonstrates SME Bank's commitment to enhancing SME competitiveness and preparing businesses for a more digital and resilient future.

Key Features of the ELSA App:

- ❖ **AI-Driven Business Diagnostics:** Uses AI to analyse SME performance and highlight strengths, weaknesses, and improvement areas.
- ❖ **End-to-End SME Lifecycle Support:** Provides assessment, advisory, training, financing access, and market opportunities in one platform.
- ❖ **Mobile-First Real-Time Insights:** Delivers instant business analysis and recommendations through an easy-to-use mobile interface.
- ❖ **Integration with ModalNiaga:** Allows SMEs to apply for financing digitally and track application status in real time.
- ❖ **Integration with ScoreXcess:** Speeds up credit scoring and assessment, reducing waiting time for approval.
- ❖ **Auto-Filled Financing Applications:** Uses diagnostic data to automatically populate forms, making applications faster and more accurate.
- ❖ **Learning and Development Modules:** Offers training programs, knowledge resources, and support tools to help SMEs grow.

References:

SME Bank (2025). *SME Bank Unveils Elsa App to Propel 10,000 SMEs Forward, Advancing Malaysia's Digital Transformation as ASEAN Chair 2025*. Retrieved from: <https://www.smebank.com.my/w/sme-bank-unveils-elsa-app-to-propel-10-000-smes-forward-advancing-malaysia-s-digital-transformation-as-asean-chair-2025-1>

Malaysia SME (2025). *SME Bank Launches ELSA App to Empower 10,000 SMEs with AI-Driven Growth*. Retrieved from: <https://www.malaysiasme.com.my/sme-bank-launches-elsa-app-to-empower-10000-smes-with-ai-driven-growth/>

Answer the following questions based on the above-mentioned scenario.

AMIT3353 MOBILE APPLICATION DEVELOPMENT**Question 1**

- a) In today's emerging market, building an app that caters to all users is essential because mobile phone hardware varies widely. An app must be optimised to run smoothly across both high-end and low-end devices to ensure consistent performance for all users.
- (i) Interpret **FOUR (4)** key challenges in developing the ELSA mobile app. (12 marks)
- (ii) Beyond mobile applications, discuss how AI technology is implemented in the ELSA app. (4 marks)
- b) In Jetpack Compose, delivering a smooth UI experience is vital for maintaining user satisfaction, encouraging engagement, and contributing to the success of a mobile application.
- Apply **THREE (3)** methods that can be used to optimise layout performance. (9 marks)

[Total: 25 marks]

Question 2

- a) User feedback within a mobile app is crucial for a positive experience. When a user completes a significant action, immediate and clear confirmation provides assurance and builds trust.
- Demonstrate **TWO (2)** methods on how the ELSA mobile app updates users after the financing application submission is completed. (6 marks)
- b) Activity instances of an app transition through different states in their lifecycles. The Activity class provides callbacks that enable the activity to detect state changes.
- Use a **diagram** to explain the activity lifecycle in the context of an Android mobile application. (9 marks)
- c) An *Intent* is an essential messaging object that acts as the primary communication method for starting new app components and navigating between them.
- Explain **TWO (2)** Android intent types with an example for each type in relation to the ELSA mobile app. (10 marks)

[Total: 25 marks]

Question 3

- a) An asynchronous task is defined as a computation that runs on a background thread and whose result is published on the UI thread.
- (i) Define the terms *UI Thread* and *Background Thread*. (4 marks)
- (ii) Provide **TWO (2)** asynchronous tasks performed by ELSA mobile app with an explanation. (6 marks)

AMIT3353 MOBILE APPLICATION DEVELOPMENT**Question 3 (Continued)**

- b) Identify the suitable data saving method (**Shared Preferences**, **Data Files** and **SQLite**) for the given data stored by the ELSA mobile app, with a simple justification.
- (i) Training Progress (3 marks)
 - (ii) Preferred Language Used (3 marks)
 - (iii) Business Registration Documents (3 marks)
- c) Location-based services are unique features of a modern mobile app, allowing it to determine the device's geographical location. This capability enables the ELSA mobile app to detect the user's current location and provide contextual, location-related information.

Differentiate the **TWO (2)** sources (Wifi / Cell Towers and GPS) to retrieve location data in terms of accuracy, speed and power consumption. (6 marks)

[Total: 25 marks]

Question 4

- a) Mobile phone cameras are versatile tools used by a wide range of applications, from basic photography to advanced features like augmented reality and video conferencing.
- (i) Illustrate **TWO (2)** manifest declarations that need to be taken into consideration in the ELSA mobile app to allow users to use mobile phone camera. (6 marks)
 - (ii) Show **TWO (2)** scenarios in which the ELSA mobile app can utilise the camera. (4 marks)
 - (iii) Choose **TWO (2)** other sensors that the ELSA mobile app can utilise to enhance user experience and satisfaction. (6 marks)
- b) Mobile app packaging is the process of preparing an application for distribution and deployment.
- (i) Justify **TWO (2)** reasons that *App Bundles* are a more suitable distribution method to distribute the ELSA mobile app to SMEs. (4 marks)
 - (ii) Describe a suitable monetisation model for the ELSA mobile app. (5 marks)

[Total: 25 marks]